

Thermal Stress Analysis and Design Guidelines for Through Silicon Via Structure in 3D IC Integration

Yuanxiang Zhang
College of Mechanical Engineering
Quzhou University
Quzhou, China
jarson_zhang@hotmail.com

Jiankun Wang
College of Mechanical Engineering
Quzhou University
Quzhou, China
2796273875@qq.com

Sijia Yu
College of Mechanical Engineering
Quzhou University
Quzhou, China
1318619415@qq.com

Abstract—Through Silicon Via (TSV) technology has been a preferred and promising strategy to achieve the reliable interconnection for the integration of 3D IC. The temperature made some changes within the procedures of the manufacture of TSV and the adoption of chips because of the mismatch in the Coefficient of Thermal Expansion (CTE) of the materials adopted within TSV structure, important thermal stress is going to be induced under the thermal load. The stresses are possible to cause different reliability problems. Dimension parameters are going to influence TSV's thermal behaviors. For the purpose of optimizing the design of TSV, Cu-filled TSV's numerical model had been set up for simulating and analyzing the influence of the thickness of SiO₂, aspect ratio (AR), diameter and shape upon TSV thermal stress within the paper. The outcomes of the simulation present that the maximum of the areas of equivalent stress always happens at the bottom and top of the interface of Cu/SiO₂. The aspect ratio's influence on the equivalent stress is rather small when aspect ratio is more than 6 for vertical TSV. Vertical TSV with thicker SiO₂ layer may improve its reliability. However, the SiO₂ layer thickness does not have obvious influence for the inclined TSV upon equivalent stress.

Keywords—Through silicon via; Finite element method; Thermal stress, Dimension parameters

I. INTRODUCTION

In recent, people have improved 3D packaging technique with high density for the purpose of reducing the size and improving semiconductor devices' performance. The signal communication among stacked semiconductor chips has been faster due to TSV technology (Through silicon via). The signal's transmission delay time that is dependent on the length of the circuit had been decreased through the comparison with the current wire bonding strategy. On the basis of 3D IC packaging, TSV mostly contains chip stacking, wafer thinning, via filling and via formation and Cu is one of the materials which are adopted most commonly for filling TSV [1-2]. During the procedure of manufacturing TSV, SiO₂ layer has been deposited upon the sidewall for the purpose of isolating via metals when the silicon substrate has been etched with a deep reactive ion etching strategy [3]. The layer is able to keep from the electrical interconnection between the neighboring silicon and the filled metal. Nevertheless, the SiO₂ layer's thermal insulation features lead to important temperature increase under high existing load. The big CTE mismatch between copper via and SiO₂ layer is also able to gather thermal stress that is going to influence TSV structures' reliability. Moreover, TSV radius as well as the ratio of the thickness of SiO₂ layer ought to be controlled in a careful way for keeping the parasitic capacitance. Therefore, thermal mechanical reliability of TSV structure is a key issue for the development of 3D integration technology.

A lot of researches have been done on TSVs' simulated thermo-mechanical behaviors [4-8]. W. Feng et al. [9] put forward an original structure TSV (Annular-Trench-Isolated TSV: ATI TSV) for the purpose of reducing the thermal stress within silicon substrate. Y. Yuan et al. [10] explored the effect of TSV's material and size on thermomechanical stress and discovered the commonly-used materials' optimal combination. Several literatures which talks about the TSV's thermo-mechanical features within the annealing procedure [11,12].

TSV structures' thermomechanical reliability within the paper has been studied through finite element method (FEM) on the basis of ANSYS multi-physics simulation platform. Moreover, the geometrical parameters of TSV shape, diameter, aspect ratio and SiO₂ thickness are considered to study their effects on the thermal stress. Comparisons among various kinds of parameters have also been shown within the paper.

II. ANALYSIS MODEL

For the purpose of analyzing the effect of a TSV's shape upon the non-linear thermal stress, two TSV shapes have been considered by adopting thermo-mechanical finite element simulation software ANSYS. The two TSV shapes (inclined and vertical TSV) have been presented within Figure 1. The thickness of SiO₂ (T), the TSV's height (H) as well as the TSV's diameter (D) have been different for the purpose of evaluating their influence on the TSV which induces stress in a thermal way.

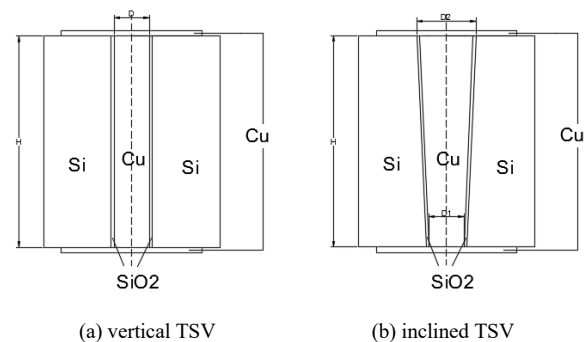


Fig. 1. Two different shapes of TSV

Simple axi-symmetric models have been adopted for various kinds of TSVs structure. The thermal deformation as well as the equivalent stress had been simulated with a ramp temperature up from 25 °C to 150 °C. Meanwhile, the whole model has been regarded heat. The stress-free temperature had been taken to be 25°C for the TSV structure, which ignores the residual stress's effect. The material features adopted within the model have been listed within Table 1.

Every material within the simulation have been regarded as isotropic.

TABLE I. MATERIAL PROPERTIES

	Young' Modulus (GPa)	Poisson Ratio	CTE (ppm/k)	Yield Strength (MPa)
Si	130.5@200K	0.28	1.4@200K	/
	128.9@300K		2.62@300K	
	128.9@400K		3.24@400K	
SiO ₂	70	0.16	0.6	/
Cu	132.9@200K	0.34	15.2@200K	250
	128.9@300K		16.7@300K	
	124.5@400K		17.5@400K	

III. FEM RESULTS AND DISCUSSIONS

A. Effect of TSV shape

Because of the great different point within the CTE between SiO₂ and Cu, the maximum stress of TSV with equivalence has been located at the interface of Cu/SiO₂. The equivalent (von-Mises) stress's distribution for a vertical via with 0.8 μ m layer of insulating SiO₂ and H=120 μ m, D=20 μ m at the temperature of 150 °C has been presented within Figure 2. The model's maximum equivalent stress is 279MPa and the via has been in the expansion condition. The equivalent (von-Mises) stress's distribution for a inclined via with 0.8 μ m layer of insulating SiO₂ and D₁=20 μ m, D₂=30 μ m, H=120 μ m is shown in Fig. 3. Compared with Fig. 2 and Fig. 3, it's obvious that the maximum thermal stress's locations are possibly the same, and the inclined via has a little lower thermal stress.

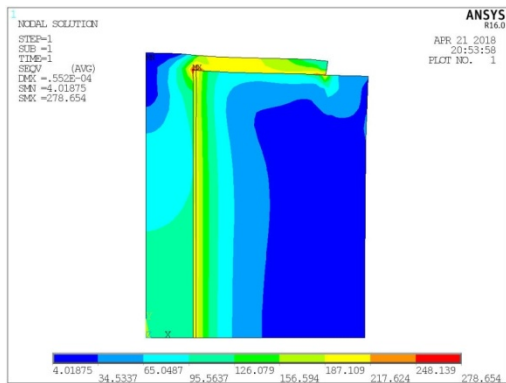


Fig. 2. The equivalent stress's distribution within the vertical TSV

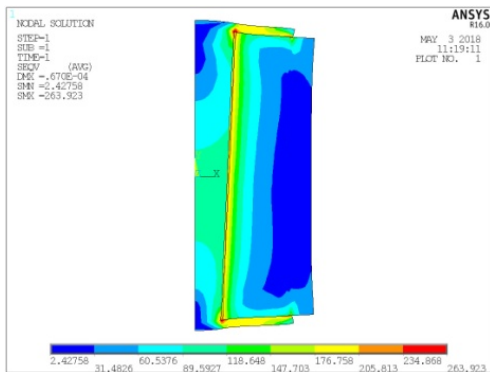


Fig. 3. The equivalent stress's distribution within the inclined TSV

B. Effects of TSV diameter and aspect ratio

For the purpose of investigating the influence of aspect ratio and TSV diameter on thermal stress of vertical TSV model, FE models whose diameter varies from 20 μ m to 60 μ m and the aspect ratio varying from 2 to 7 have been simulated which are presented within Figure 4. The thickness of the SiO₂ layer has been kept at 0.8 μ m. The boundary condition stays the same with the initial analysis. Fig. 4 presents the equivalent stress within various kinds of diameters and aspect ratios. According to the figure, it's obvious that the equivalent stress reduces greatly with the TSV diameter's increase. Thus, the TSV's diameter has been a significant design parameter in manufacturing processes. The equivalent stress also reduces with the aspect ratio's increase when the diameter has been constant. However, the equivalent stress is possible to keep unchanged with the aspect ratio's increase when it's over six. It shows that the aspect ratio does not have any obvious influence for vertical TSV on the equivalent stress.

Fig. 5 shows the equivalent stress in via height in the inclined TSV with D₁=20 μ m, D₂=30 μ m. It's obvious that the maximum equivalent stress first reduces quickly and later keeps stable with the via height's increase. The TSV diameter also has obvious influence on the equivalent stress within the inclined TSV. The equivalent stress increases with the diameter's increase when D₂/D₁=2 which is presented within Figure 6.

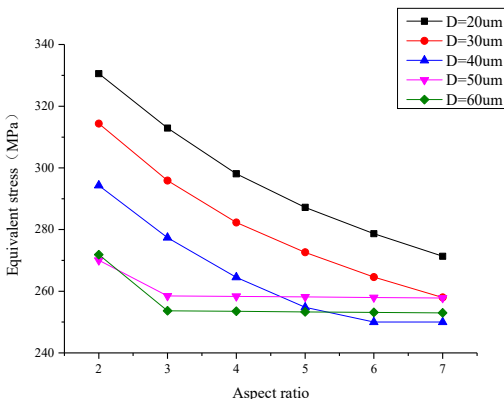


Fig. 4. The equivalent stress within various aspect ratios and diameters for the vertical TSV

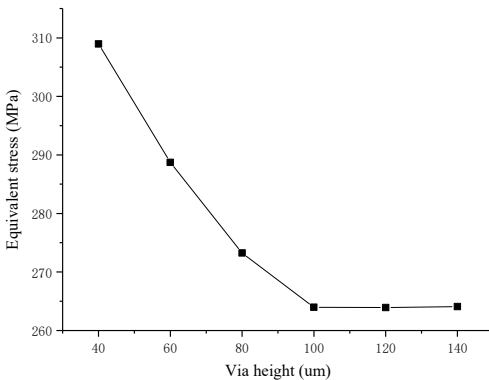


Fig. 5. The equivalent stress in via height for the inclined TSV

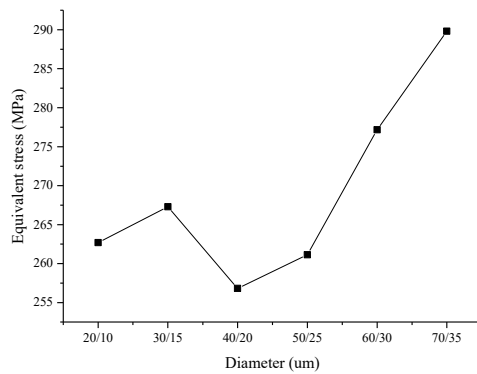


Fig. 6. The equivalent stress for the inclined TSV within diameter

C. Effects of SiO₂ thickness

The relation between the thickness of SiO₂ and the equivalent stress for the inclined and vertical TSV have been presented in Figure 7 and Figure 8. For vertical TSV, as the SiO₂ layer thickness grows from 0.6 μm to 1.4 μm, the maximum von Mises stress reduces from 284 MPa to 271 MPa. Thus, TSV which has thicker SiO₂ layer is possible to develop its reliability. Nevertheless, the thickness of the SiO₂ layer does not have any obvious influence for the inclined TSV on the equivalent stress.

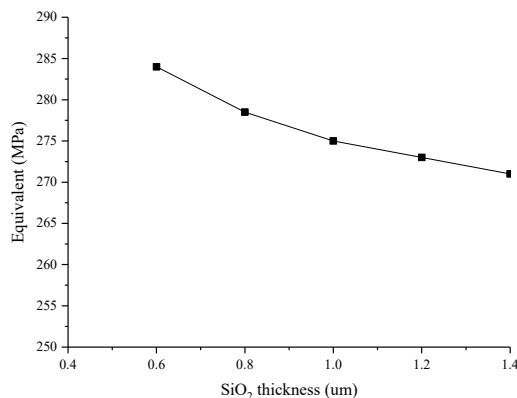
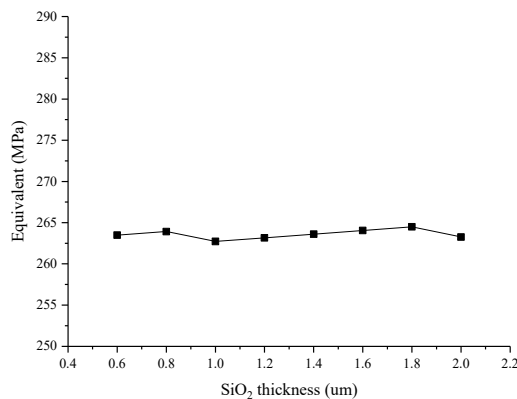


Fig. 7. The equivalent stress in SiO₂ thickness for the vertical TSV



The equivalent stress in SiO₂ thickness for the inclined TSV

IV. CONCLUSIONS

The analysis of the non-linear thermal stress of TSV within the paper with various kinds of shapes, diameters, aspect ratio and SiO₂ thickness have been explored. Several significant conclusions have been summed up below.

(1) In terms of every TSV discussed, the maximum stress areas with equivalence always appear at the bottom and top interface of Cu/SiO₂.

(2) For vertical TSV, the equivalent stress is possible to reduce with the aspect ratio's increase when the diameter has been constant. However, the equivalent stress may have no changes with the aspect ratio's increase when it is over six. For inclined TSV, the maximum equivalent stress first decreases rapidly and then becomes stable with the increase of the via height.

(3) Vertical TSV with thicker SiO₂ layer may improve its reliability. However, the SiO₂ layer thickness does not have any obvious influence for the inclined TSV on the equivalent stress.

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REFERENCES

- [1] M. Motoyoshi, "Through-silicon via (TSV)," *Proceedings of the IEEE*, Vol. 97, Nos. 1, pp. 43-48, 2009.
- [2] J. H. Lau, "Evolution and outlook of TSV and 3D IC/Si integration," In *IEEE International Conference on Electronic Packaging (ICEP)*, Nara, Japan, pp. 13-15, April, 2010.
- [3] P. Leduc, M. Assous, Di Cioccio L., et al., "First integration of Cu TSV using die-to-wafer direct bonding and planarization," In *IEEE International Conference on 3D System Integration*, pp. 1-5, 2009.
- [4] J. Zhou, D. Yu, R. He, F., et al., "Nonlinear thermal stress & strain analysis of through silicon vias with different structures and polymer filling," in *2011 IEEE 13th Electronics Packaging Technology Conference*, 2011, pp. 686 - 690.
- [5] Y. Pan, F. Li, H. He, J. Li, and W. Zhu, "Effects of dimension parameters and defect on TSV thermal behavior for 3D IC packaging," *Microelectronics Reliability*, vol. 70, pp. 97-102, 2017.
- [6] J. Hwang, S. Seo and W. Lee, "Effect of Design Parameters on Thermomechanical Stress in Silicon of Through-Silicon Via," *Journal of Electronic Packaging*, vol. 138, p. 031006, 2016.
- [7] M. Xiao, F. Wu, H. Liu, and W. Xia, "Electric-thermo-mechanical analysis of TSV under high current stressing," in *14th International Conference on Electronic Packaging Technology*, 2013.
- [8] T. F. Jiang, J. Im, R. Huang, and P. S. Ho, "Through-silicon via stress characteristics and reliability impact on 3D integrated circuits," *MRS BULLETIN*, vol. 40, pp. 248-256, 2015.
- [9] W. Feng, N. Watanabe, H. Shimamoto, K. Kikuchi, "Analysis of thermal stress distribution for TSV with novel structure," in *2014 International 3D Systems Integration Conference (3DIC)*, 2014, pp. 1046-1053.
- [10] Y. Yuan, L. Gao, W. U. Hao, L. I. Ming, "Influences of size and material of through-silicon via on thermo-mechanical stress," *Semiconductor Optoelectron*, vol. 34, no. 2, 2013, pp. 255-258.
- [11] W. S. Kwon, D. T. Alastair, K. H. Teo, et al., "Stress evolution in surrounding silicon of Cu-filled through-silicon via undergoing thermal annealing by multiwavelength micro-Raman spectroscopy," *Applied Physics: Letter*, vol 98, 2011, pp. 232106.
- [12] H. A. S. Shin, B. J. Kim, J. H. Kim, et al., "Microstructure evolution and defect formation in Cu through-silicon vias (TSVs) during thermal annealing," *Journal of Electronic Material*, vol. 41, no.9, 2012, pp. 712-719.